
ECE228 Project Proposal:

Assessing the Impact of Temporal Attention Mechanism on the End-to-End Self-Driving Models

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1 Problem Background & Motivation

1.1 Background

1.1.1 Shift From Modular to End-to-End Self-Driving

Traditional self-driving system typically rely on a modular architecture, separating the task of driving into several parts, such as sensing system, mission planning layer, behavior layer, and so on(1). Though the components are highly interpretable, these modules are susceptible to error compounding. A minor misclassification in the sensing module would cascade into a catastrophic failure in the final output. Meanwhile, defining every physical rule and edge case in the complex driving environment is exhaustive in computation(2). To address this, End-to-End self-driving emerged as a biologically inspired alternative. The End-to-End system treat self-driving as a supervised regression problem. It trains a unified neural network to map sensor inputs directly to low-level physical control instructions, ignoring intermediate representations.

1.1.2 Baseline:Pure CNN Architecture

The End-to-End self-driving method, proposed by NVIDIA, utilized a pure CNN trained on minimum driving datasets(3). The model successfully learned to extract spatial features, such as lane mark and road edges and output continuous steering angles. It even operated in areas with unclear visual guidance. Although the method has been successful in lane-keeping tasks, it suffers from a fundamental physical limitation: it is memoryless. The pure CNN model processes video sequences under the assumption of independent and identical distribution, treating each frame as an isolated entity. However, in reality, the car's movement state at time t is strictly dependent on its state at time $t-1$. Without temporal continuity, pure CNN model is highly fragile to physical noise, such as blocking the camera for a short time.

1.1.3 Introduction of LSTM

To fix CNN model's spatial-only defect, later researches focused on integrating temporal information into the End-to-End method. In 2017, researchers proposed an architecture combining FCN with LSTM(4). In this spatial-temporal architecture, FCN acts as a spatial encoder to extract low-dimensional features from individual frames, while LSTM serves as a temporal decoder to maintain a hidden state that propagates historical movement states. This combination allows the model to learn high-level physical parameters such as velocity and angular acceleration, resulting in smoother and more mechanically realistic steering trajectories.

1.2 Motivation

Though the evolution from pure CNN to CNN-LSTM combination architecture effectively addressed the memoryless defect, a critical problem remains, regarding the reliability of sensors. Standard

LSTM treat all historical inputs with a sequential weight mechanism, which lack the ability to assess a individual frame's quality. Thus, when sensors meet with sudden physical degradation such as sun glare, heavy rain drop, the LSTM's hidden state will be heavily influenced by these noise frames, resulting in producing unreliable steering instructions.

Motivated by the attention mechanism, this project propose that not all historical frames contribute equally to the output of driving instruction, particularly under extreme sensor noise. By integrating a temporal attention mechanism, the model will gain the ability to assign different weights to different frames on the time step based on their feature clarity and the time elapsed. This enables the model to actively suppress the influence of degraded frames.

2 Related Work

In the paper *End to end learning for self-driving cars* by Bojarski, M., et al., they pioneeringly proposed the method to train a CNN to directly map frames to steering angle, avoiding the complex modular systems. Based on that, *End-to-end learning of driving models from large-scale video datasets* by Xu, H., et al. introduced LSTM to capture the temporal dependencies, successfully making the model remembers the historical frames. On top of that, *Interpretable learning for self-driving cars by visualizing causal attention* by Kim, J., & Canny, J. was the first to utilize attention mechanism in self-driving, but more in spatial domain(5). This project, on the other hand, applies attention to the temporal domain attempting to solve the problem of sensor noise.

3 High-level Methodology

This project proposes a method, attention-enhanced temporal network, aiming to improve the End-to-End self-driving model's robust against sensor failures. To realize that, a temporal attention module will be injected in between the CNN-LSTM model, which will calculate an attention score for every historical frame based on its feature quality. This allows the model to automatically suppress the influence of unreliable frames and aggregate context only from reliable historical states. To validate the method, a custom data-loading pipeline that contains deliberately injected physical noise will be constructed to compare the model's performance with the baseline model(CNN+LSTM).

References

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